

ATN/IPS SECURITY APPROACH: TWO-WAY MUTUAL AUTHENTICATION, DATA INTEGRITY AND PRIVACY

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Abstract

Currently there are various industry and regulatory activities (Airlines Electronic Engineering Committee Internet Protocol Suites (IPS) Working Group, Radio Technical Commission for Aeronautics SC-223, EUROCAE Working Group 108, and ICAO Working Group I) underway to develop the future aviation safety service Aeronautical Telecommunication Network (ATN) based on Internet Protocol Suite (IPS) IPv6, which is the selected protocol for air/ground communication in support of Air Traffic Service (ATS) safety service applications. This will provide IPv6 as an alternative to the traditional ACARS and ATN/OSI protocols for the air/ground aviation safety services network, but it also exposes the aviation safety services data communication to various cyber security threats. Data communication security plays an important role in the successful development of ATN/IPS as a next generation aviation safety service network that support ATS, ATC, AOC messages over various air/ground radio links. Security countermeasures help ensure the confidentiality, availability, and integrity of ATN/IPS systems by preventing or mitigating harm from cyber security attacks. This paper introduces the motivation and context for ATN/IPS security in terms of the aviation safety service data communication network. It describes the first fully implemented two-way authentication security, data integrity scheme for the aircraft air/ground safety service communications based on existing internet standards, specifically the Datagram Transport Layer Security (DTLS) protocol. The security approach uses Elliptical Curve Cryptography (ECC), which is the most efficient, matured, and widely accepted public key cryptography algorithm. This paper presents the DTLS implementation in the context of an air/ground system architecture and overall feasibility and the scheme's feasibility (low overhead and high interoperability). This is further demonstrated through extensive evaluation of a prototype using an existing avionics hardware platform and an existing ground system.

1. Introduction

The traditional Aircraft Communications Addressing and Reporting System (ACARS) has been used in air/ground communication between aircraft and the ground for well over 30 years. Ever since airlines first used ACARS in 1978 to carry Aeronautical Operational Control (AOC) messages and specifically to assist with the reporting of aircraft OOOI (Out of the gate, Off the ground, On the ground, and Into the gate) events in the airline industry [1]. Since the early days, the application of ACARS-based messaging has increased rapidly in size and quantity. Currently ACARS is used worldwide for aircraft position and status reporting to airline back offices and airline operational management, as well as to Air Traffic Control (ATC). Europe has deployed an ATN network using the OSI protocol (ATN/OSI) for ATC use. Both communication technologies (ACARS and ATN/OSI) are well-known protocols offering limited security, however the lack of Internet connectivity, the single-purpose protocols, and the specialized communication media perhaps make ACARS and ATN/OSI less likely to attract bad actors.

Currently there are various industry and regulatory activities (AEEC IPS Working Group, United States RTCA SC-223, European EUROCAE WG-108, and ICAO WG-I) under way to develop aviation standards to migrate air/ground safety service communications from ACARS and ATN/OSI to the Internet Protocol Suite based aviation safety service Aeronautical Telecommunication Network (ATN/IPS). Internet Protocol Suite IPv6 is the selected protocol for the air/ground data communications to become an alternative to ACARS and ATN/OSI, but it exposes the aviation safety services data communication network to various cyber security threats since internet protocols bring many security vulnerabilities with it. Bad actors could attempt attacks on aircraft systems using ATN/IPS. Once ATN/IPS is widely used, the consequences of an advanced persistent threat would

be dire. ATN/IPS network could become congested, airline operations chaotic, and safety critical systems in the aircraft and on the ground could be affected by threats using the IPS directly or indirectly. As shown in Figure 1, the IPS aircraft is a multi-homed system having multiple interface with different access networks and service providers. There is a strong security need to protect the aircraft within each access network and between the aircraft and each service provider.

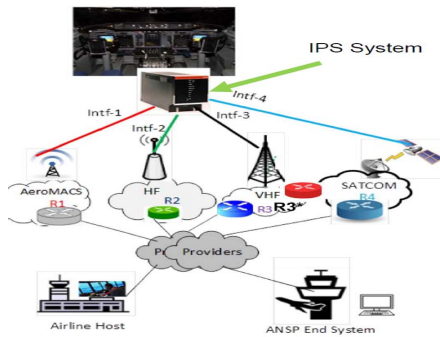


Figure 1. ATN/IPS Entities Including Aircraft and Ground End Systems

As shown in Figure 2, when the air/ground communication uses an Internet Protocol based network, it will be one of the largest mobile IP networks, since a node (aircraft) moves across multiple continents and each continent will have a large number of aircraft. Because of this global scope, security becomes a tremendously important issue. In mobile ATN/IPS networks, mobile nodes join or leave the networks frequently and the network topology is constantly changing, it is also challenging to manage the network. In this mobile environment, where the network topology is constantly changing, multipath routing is vulnerable. Challenges to an IP based network may include malicious actors, who may fabricate packets in order to disrupt communications because of a lack of integrity and security checks [33], [34]. Without security and integrity, all users would be potentially vulnerable to misuse of the services provided using ATN/IPS. Hence, it is necessary to deny access to compromised, defective, and illegitimate nodes. This requires mutual authentication between a mobile node (aircraft) and the network.

Figure 3 depicts the end to end communications for the existing air/ground system and the context of how IPS is incorporated. Many current avionics

designs do not seamlessly support application security. Historically, stand-alone avionics systems were proprietary implementations. A large number of independent legacy avionics systems co-exist across the aviation network, but have no security functions. Their silo nature conflicts with the open spirit of the internet protocol and hence introduces problems of interoperability and service-level interaction, which limits the benefits of internet protocol based systems, and could impede large-scale IPS deployment.

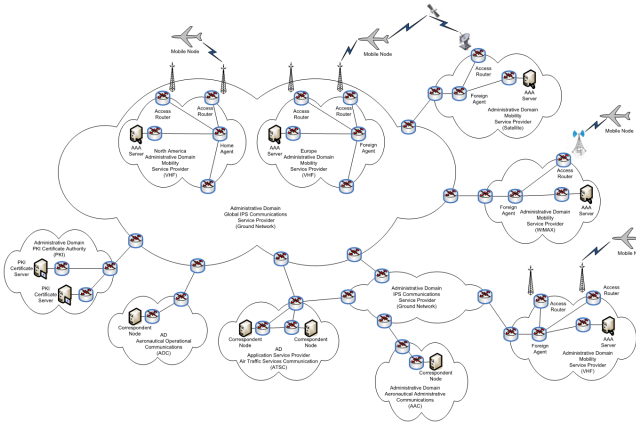


Figure 2. IPS Network Topology

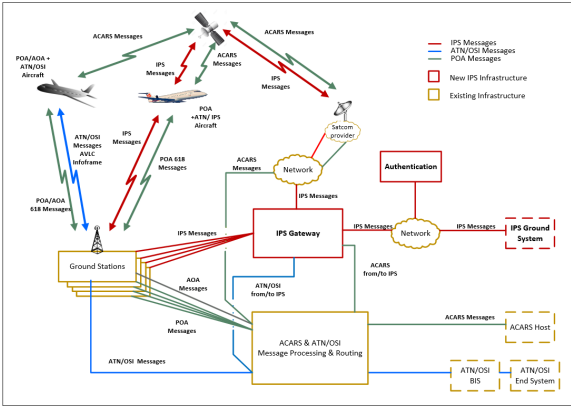


Figure 3. Air to Ground Communication with IPS

An ATN/IPS architecture with a ground message gateway is proposed in an attempt to unify ATN/IPS solutions while supporting existing systems. Further, this solution does seamlessly support mobility, which is a major requirement of the ATN/IPS and a desirable characteristic for AOC and ATS applications. In our approach, we proposed a message gateway framework at the IPS network that can provide service to legacy ACARS and ATN/OSI

end systems, while supporting air/ground IPv6 communication. Data from aircraft avionics and customer end systems is processed through the IPS message gateway when at least one endpoint is IPS enabled. The message gateway can support IP to IP end system, IP to an ACARS end system and IP to an ATN/OSI end systems. This not only supports the future IP based network, it will also allow IPS aircraft or ground systems to communicate with legacy ACARS or ATN/OSI aircraft or ground systems. This will support an extended transition period to facilitate a gentle transition to an entirely IP based network. The IPS message gateway is media independent, initially it will support VDL mode 2 and Satcom but other media can be added as needed as shown in Figure 3.

Providing the network security is an important objective in the design and implementation of the IPS message gateway. For the IPS traffic, the basic requirements are to support mutual authentication, data integrity and provisions to support privacy or data confidentiality. To protect the existing VDL M2 link, we have proposed the using DTLS [21] over connection-less AVLC (AVLC UI Frame). The DTLS provides the mechanism for key exchange and to dynamically negotiate session keys. DTLS provides the required security functions: key exchanges, session management, two way mutual authentication, and data integrity between the ground system and the aircraft. We are introducing the first fully implemented two-way authentication security scheme for the aircraft to ground safety services. Our Implementation is based on existing Internet standards, specifically the DTLS protocol. By relying on an established standard, existing implementations, engineering techniques and security infrastructures, IPS can use a strong security posture that is straightforward to implement in the air and on the ground.

We propose the provision to encapsulate AOC, B1, B2 and FANS 1/A air/ground traffic using ATNPKT [29]. Message exchanges between IPS aircraft and the IPS gateway will be encapsulated in the ATNPKT. We propose the provision to use DTLS to provide the mutual authentication, key exchanges and data integrity between the aircraft and the message gateway independent of the air to ground media. This framework combines different security mechanisms and providing a global security solution defined in this paper. The basic solution envisioned

to achieve the security requirements is to use DTLS between an aircraft and ground systems with linked digital certificates (generated by a trusted third party). These certificates can then be used to digitally sign information. Most of the existing solutions manage these certificates by means of a central Certification Authority (CA) [3]. ATN/IPS networks will rely on the public key infrastructure (PKI). In PKI a CA issues an authentic digital certificate for each node in the network. Aircraft and ground will use digital certificates to authenticate each other. Therefore, efficient certificate management is crucial for the robust and reliable operation of any PKI. A critical part of any certificate management scheme is the revocation and management of certificates.

The remainder of this paper is organized as follows: Section 2 identifies the process of establishing a secure session. Section 3 provides the security and risk analysis. Section 4 highlights high level security requirements. Section 5 provides methods for mutual authentication and data integrity. Section 6 describes the PKI and certificate management framework. Section 7 specifies a DTLS-based secure aircraft communication path and proof of concept implementation. Section 8 provides the results from the prototype followed by findings in section 9. Finally, the paper concludes with Section 10.

1.1 Related Work

A vast amount of related literature can be found on the topic of network security in general [11], [14], [25], [31], [32] and of security for wireless networks [13], and [20] on over all network security . Hansman et al present a four dimensional approach to attack taxonomy, including target, attack vector, exploit, and vulnerabilities [14]. Basin et al [30] presents the security vulnerabilities in the embedded system in various protocol layers. As an excellent entry point, [38], [39] provides an extensive overview on the security needs of IP Network communication. Three aspects shall be examined in more detail, as they are crucial with regard to the security requirements. They are mutual authentication, data integrity, and privacy. Other aspects, like secure routing are also relevant for the anticipated application scenarios.

There has been quite significant progress in the security architecture on embedded ECC public key

cryptosystems and as a result of this progress DTLS can be used in aviation. Embedded internet ECC public key cryptosystem use is described in [40] paper. IETF provides the profile to the use of DTLS on resources constrained environment [36] based on the security architecture networks on DTLS.

In paper [26], one of the main criteria for session key agreement is provision of forward secrecy. The forward secrecy means that disclosure of long-term secret keys does not lead to exposure of session keys from earlier runs [5]. Mutual authentication is described in the Davie [8] scheme. In his scheme, when mutual authentication is realized between the user and the server, they agree on a session key to be used for protecting their subsequent communications.

Buttayan, Ravi et al [5], [18] present the various security threats related to embedded systems. There has been extensive research done on mitigating the effect of IP spoofing. A similar approach was proposed by Duan et al [7] by introducing the concept of the feasible upstream router and deeming any packet that does not originate from such a router a spoofed packet. Wang et al [10] presents the secure routing protocol provision.

In this respect, there are more formalized approaches and these provisions should be considered in data communication security. This includes mutual authentication, data integrity, non-repudiation and privacy [14], [15]. Astatke et al [15] describes the method to use public key cryptography in wireless sensor networks using Elliptic curve cryptography (ECC) as public key cryptography scheme and advanced encryption system (AES).

ATN/IPS requires an efficient way to enable end to end secure communications. DTLS which is used to secure UDP communication is the transport protocol recommended by the IETF for resource constrained devices [21]. ECC cipher suites in the RFC [36] such as CipherSuite TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SH A384 provides security that considered reasonably future proof. ECC is used to generate key pairs, and provide entity authentication, message integrity and privacy. This is used in order to create the key pair used in asymmetric cryptography that is more efficient than the RSA and DSA. It is possible in fact to obtain the same strength with less bits composing the keys. This is applied in both authentication and key exchange [22] purposes. Ephemeral Elliptic

Curve Diffie-Hellman (ECDHE) algorithms provides the forward secrecy and key exchange and mutual authentication with DTLS [21].

In this paper, the DTLS with ECC CipherSuite TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SH A384 provides the authentication, message integrity and privacy for the future ATN/IPS Internet architecture as the basis for our aviation air to ground communication. We note, though, that some of our security mechanisms are general and can be applied to other future Internet based systems.

2. Security Analysis

Figure 4 depicts the air to ground communication, assets with potential attack paths and threats. The aviation network is vulnerable to multipath attacks due to multiple air to ground links. Attackers can flood very high frequency data radio (VDR) with data by launching Denial of service (DoS) attacks [35], [33] or corrupt or modify messages. In DoS attacks, Intruders send a flood of packets preventing the system from performing normal communication. An avionics should drop such intruder messages and let only genuine message pass through. The VDR, an avionics embedded device, is always at edge of the network, and when there is flood of information requests that come to it, its performance is impacted. Deep packet Inspection is one of proven defense techniques against DoS. Deep packet Inspections is a technology that inspects each packet, based on set of rules, only allows packets that passes all rules to pass through the network. Systems that execute deep packet inspection require fast processors to inspect each packet without degrading system performance. Usually the air to ground radios do not have those processing resources so it is performed by the ATN/IPS message router level. Other threats include message corruption and spoofing, which will require source authentication, message integrity and privacy to protect the air to ground message.

The ATN/AIPS extends the Internet to the aircraft's safety service world and thus poses many new security and privacy challenges. Some of the problems are due to the intrinsic characteristics of the avionics and its differences compared to traditional networks, while others arise because of the integration of the aeronautical safety services with an Internet protocol based network. As shown in Figure

4, bad actors may come in at different points to attack ATN/IPS systems. To protect against those attacks, it is important to examine the security problems according to the information flows and potential entry points. Security problems are listed below:

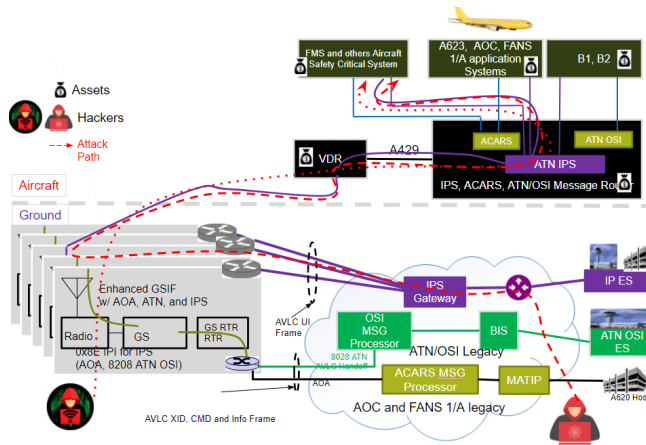


Figure 4. IPS Attack Path and Threats

Integrity: Mobility and wireless communication links for avionics devices raises concerns about data integrity. Once deployed, most of these devices will operate with minimal or no maintenance and are expected to exhibit a long mean time between failures. With very limited maintenance, cyber tampering is a much easier task than in a supervised wired network. Further, as a result of open and wireless link corruption or manipulation of data by an attacker, the data transmitted by avionics devices could have low integrity and might be corrupted at the environmental level [31]. Avionics data without integrity checks may be easy to spoof and forge.

Authentication: The avionics safety service network is a distributed deployment of a large number of avionics devices and ground systems, such as airline hosts, ANSP end systems, vendors, weather service providers, and suppliers and will generally be deployed in various continents and without any specific protection, which makes the devices difficult to manage and vulnerable to various attacks. For example, an illegitimate node may register itself claiming that it is an aircraft at one location while it is actually at a different location. Or a ground node installed for monitoring aircraft is moved to another location by a malicious person. This introduces the challenge of authenticating avionic devices, which involves recognizing the device and verifying its association with a correct topological address.

ATN/IPS networks require some mechanism to authenticate messages, identify valid aircraft and ground entities, and arbitrate suspect ones. Because an ATN/IPS network is open in the way that mobile hosts can come and go, there is no way to know, which mobile hosts are present in the network. If some data for example are being transmitted, it is important to make sure that the communication is established with the right host.

Confidentiality: the communication method between avionics devices and the ground gateway is primarily wireless, which results in confidentiality risks. Unfortunately, unlike many other wireless environments, such as cellular and Wi-Fi networks, it is difficult for the ATN/IPS network to provide confidentiality for data transmission due to the resource-constrained nature of aviation devices, which are a large fraction of current safety service avionics devices. Different from typical devices in traditional wired and wireless networks, avionics have very limited resources and capabilities. Constraints on computational capability, storage and other aspects of an avionics device introduce a high barrier for it to perform the necessary operations to achieve data confidentiality, such as through encryption and key management.

Secrecy or Privacy: By connecting the aviation world objects and information through the Internet protocol, data may become accessible to various organizations and domains across the network, instead of only being revealed to a small group, which makes it more likely to be exposed to sophisticated malicious parties hence increases the probability of being attacked and exploited.

3. Security Requirements

The protocol for establishing mutual trust and keys described in subsequent sections can support capable devices as well as highly resource-constrained devices. Second, the establishment of the avionics device keys, which conventionally requires the protocol to leverage a trusted public key infrastructure (PKI). Traditional PKI can be challenging to establish on a global scale, and may require computational resources for cryptographic processing and substantial IT professional expertise as well. Particularly, when developing the key provisioning protocol, the following properties and requirements need consideration:

Low communication overhead: aircraft and ground communication (mutual authentication information) should be as small as possible.

Authentication: Authentication service is to verify an aircraft or ground identity and to assure the recipient that the message is from the source that it claims to be from. At the time of communication initiation, the service assures that the aircraft and ground are authentic, that each is the entity it claims to be. The service must assure that a third party does not interfere by impersonating one of the two legitimate parties for the purpose of authorized transmission and reception of messages.

Mature Technology: Technology should be matured, widely peer reviewed and adopted by industry in general.

Regional cryptographic concern and consideration: Proposed solution should be deployable to various countries with different laws and regulation.

Scalability: Many avionics systems are large scale deployments, it would be extremely difficult to establish a secure credential for each device if human intervention is required. To enhance usability and chance of practical deployment (as well as reducing human error), it is important and necessary to facilitate automatic trust establishment.

Dynamic mutual trust: For a successful protocol, each aircraft and ground entity involved in the mutual authentication can reach mutual trust at the end of the procedure. Further, the aircraft establishes the mutual trust dynamically with the ground entities and establishes the session and master secret that are used for the key derivation to provide the security.

Consistency: protocol participants should have a consistent view of the protocol session. Namely, the aircraft, ground and certificate authority.

Secrecy and forward secrecy: The principle of least privilege should be followed, implying that any resultant secret key should only be shared between the involved two parties. No third party should be able to distinguish the resultant key from a random key. Forward secrecy should be conserved so that the compromise of a long-term secret key will not affect the confidentiality of past communication sessions [6].

Low computational cost: As we focus on avionics, efficiency is a priority. Ideally, computations performed by avionics should be simple and fast. When designing the protocol, the avionics, which is typically, though not necessarily, a resource-constrained device, should be protected by shifting workload to the ground as much as possible.

Average maintenance requirements: The protocol should require as little professional knowledge as possible. The configurations are automatic and user-friendly so that people do not need to be IT experts to operate the protocol

4. Protecting VDL Mode 2

As noted earlier, VDL Mode 2 will be one of the initial air to ground media for ATN/IPS and unlike Satcom and future media, this access network is not currently protected. In order to protect the VDL M2 access network two way mutual authentication, and data integrity with privacy provision is added in the AVLC layer. Since IPv6 MTU size minimum is 1280 octets, and current AVLC networks typically support smaller MTU sizes in the AVLC UI frame, a new 'orange' protocol (shown in the Figure 5) is added to support the fragmentation of IPv6 packet in the AVLC UI frame layer.

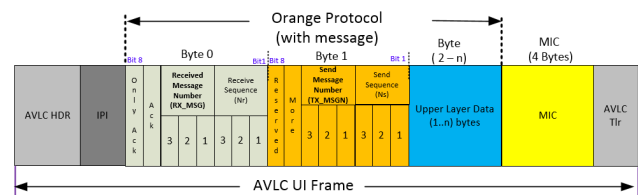


Figure 5. Orange Protocol to Support AVLC

AVLC header is documented in the ICAO DOC 9876. 0x8E value in the IPI field indicates that it is carrying IPS traffic. First octet in the upper layer data field identifies type of traffic AVLC is carrying: value of 0x0A identifies DTLS authentication traffic. When IP packet is not fragmented, orange protocol set the Ns and Send Message number both to 0 and no acknowledgement is sent for the un-frgmented frame. This is because upper layers will acknowledge the message, so acks for the single frame message would be redundant. If frame is fragmented, send message number will start at 1 with Ns range 0 to 7 and more bit is set to 1 except the last frame which indicates the last frame of the fragment.

Acknowledgment is sent for the high water mark. When aircraft receives a Ground Station Information Frame (GSIF frame) identifying IPS service, it identifies the DSP based on the system mask and if authentication has not been established, it will initiate the DTLS authentication session as shown in Figure 6.

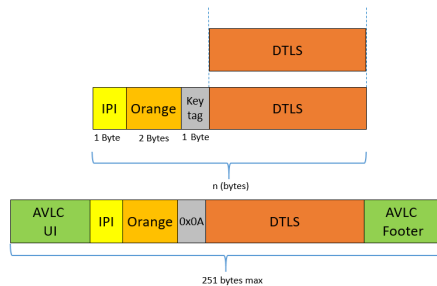


Figure 6. DTLS Authentication Session

Once the DTLS session is established, the aircraft will derive the session key for the message integrity validation. Message Integrity Code (MIC) which is ECDSA-SHA384-32 [36] is added to each AVLC frame. The MIC spans the AVLC header, orange header and data. Each message MIC is calculated, and the last 4 bytes of the computed value are added within the frame. The receiving end verifies the MIC based on the session keys and Figure 7 illustrates the AVLC UI frames with added security provision.

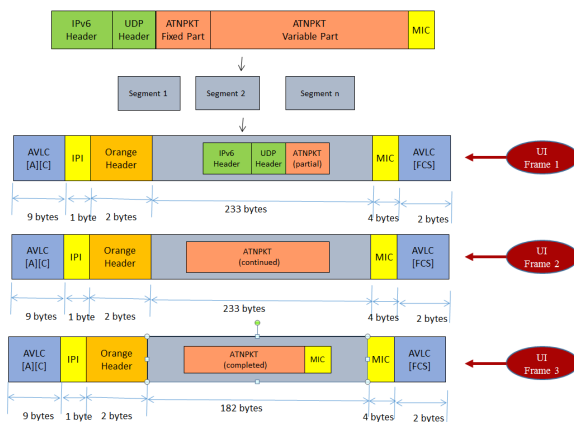


Figure 7. AVLC UI Frame with Orange and MIC

5. IPS Message Protection

As shown in Figure 3, the IPS Gateway is designed to facilitate communications with IPS

equipped aircraft using existing air-ground network infrastructure and to accommodate future air-ground links. Figure 2 shows the network topology for the mobile ATN/IPS and Figure 3 depicts the entities related to the air to ground communication in context of various air to ground links and access networks. The IPS Gateway will initially interface with IPS Aircraft using VDL Mode 2 and Satcom, with IPS Ground Systems, with legacy A620 airline hosts, and with ATN/OSI End Systems. Figure 8 identifies the interfaces and data flow that the IPS Gateway supports for IPS. The IPS Gateway will also support communication of non-IPS aircraft (ATN/OSI and ACARS) with IPS Ground Systems. Message encapsulation mechanism between air to ground for various message types are shown in the Figure 9.

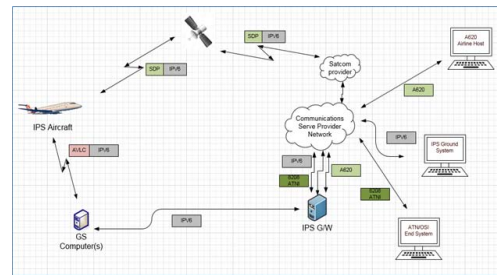
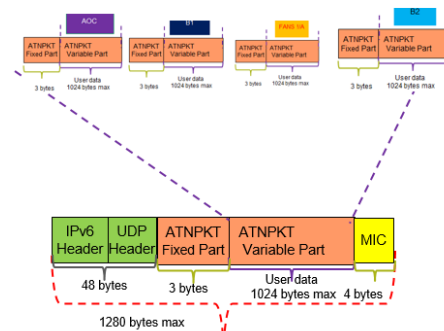


Figure 6. Data Flow to/from IPS Aircraft



The IPv6 minimum MTU is 1280. IPv6 MTU will accommodate the ATNPkt.

Figure 7. IPS Message Encapsulation

The ATNPkt [29] is the basic unit in IPS communications. There are three modes in which the ATNPkt is used: a) Session establishment message exchange, b) Session management message exchange and c) Application message exchange. Figure 10 provides service type ports that are provisioned for IPS.

The Initial Protocol Identifier (IPI) identifies the presence of IPS data and the UDP port number is

used to describe the type of IPS data. Additionally data on the authentication port (5908) has a key tag to further identify the type of message. The use of the ATNPKT for these three modes is shown below and the individual components of the protocol build-up are detailed further on in the paper.

Service Name	Port	Notes	Avionics Configuration	Ground Configuration	Current IANA Status
Authentication / Management	5908		Client	Server	Need to register
AOC	5909	ARINC 620 data	Both	Both	Need to register
CM	5910	IP App	Both	Both	Registered
CPDLC	5911	IP App	Server	Client	Registered
ADS-C	5912	IP App	Server	Client	Registered
AFN	5915	ARINC 620 data	Both	Both	Need to register
FANS ADS-C	5916	ARINC 620 data	Server	Client	Need to register
FANS CPDLC	5917	ARINC 620 data	Server	Client	Need to register
All other ports	All other	Native IP	Application specific	Application specific	Varies

Figure 8. Service Endpoint Definition

The Initial Protocol Identifier (IPI) is a one-byte field used to identify the presence of IPS data. IPI 0x8E value is identified for Ipv6 per ISO/IEC TR 9577 1999 edition appendix C [16]. The ground adds the IPI before the IPv6 header for all uplink messages. For downlink messages, the ground station (VHF or Satcom) examines the IPI and routes IPv6 messages to the IPS Gateway. The IPI will be included as a part of the message in transmission to the IPS Gateway.

ATNPKT IP Dialogue service layer initiates the authentication between the aircraft and the message gateway using the DTLS. Protocol detail and cipher suite is same as the AVLC layer which is further detailed in subsequent sections. This authentication can be omitted for the same DSP since the key can be derived based on the AVLC layer DTLS session master secret and used for the ATNPKT layer security for each DSP.

5.1 Session Establishment

The protocol build-up for session establishment (authentication) is shown for IP-based communications (example of this is shown in Figure 11); non-IP based communications adds another layer. Session establishment shall utilize

UDP port 5908. Port 5908 is reserved for specific messages (authentication, key management, IPS information, and IP lookup); with the type of message being defined by the first byte (key tag) of the UDP data field. For authentication, the key tag field value must be 0x0A. Prior to authentication, port 5908 will be the only open port. Note that a message integrity check (MIC) field is not present during authentication because the session key has not been established. No other key tags will be accepted by the gateway prior to authentication

IPI	IPv6 Header	UDP Hdr	Key	ATNPKT 9596 Fixed				ATNPKT 9596 Variable			
0x8E	src & dst addresses, etc.	src & dst ports, etc.	0x0A	Ver	DS Prim	App Tech Type	More	Presence Flags	Seq	Length	
		5908								DTLS	

Figure 9. ATNPKT Used for Authentication

5.2 Session Management

This message exchange covers key management and support messages (IPS information, IP lookup). All of these messages are also on UDP port 5908, with the specific type of message being identified by the key tag. The format is the same as session establishment except that it includes MIC field since authentication has been completed. It should be noted that only Authentication, DTLS alerts and Key Management Messages on UDP Port 5908 use the DTLS header, all messages that use a DTLS header post authentication will be encrypted. After authenticate all data in 5908 will be encrypted. For instance: if an aircraft sends an information message or an IP lookup request it would be encrypted. The IP Lookup response would be unencrypted but MICed as well.

5.3 Application Message

The application messages are sent on specific UDP ports other than port 5908. These messages do not require the key tag used for port 5908 messages. Examples of the protocol build-up are shown in Figure 12.

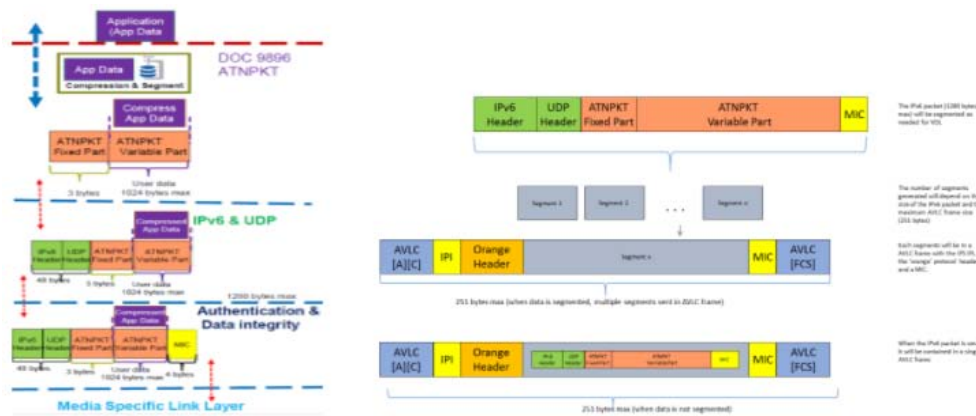


Figure 10. IPS Air to Ground Protocol Overview

6. Data Authentication and Integrity Methods

The first step for an IPS aircraft communicating with any entity is to authenticate with the IPS Gateway. Authentication is initiated by the aircraft. DTLS will be implemented for authentication in order to protect the IPS data. For DTLS, X.509 certificates with a minimal parameter set and ECC 384 bit public keys are used to mutually authenticate aircraft and ground. The client certificate is directly signed by a root CA or service provider. The exchanging of PKI keys in DER format while efficient, will likely lead to multiple fragments to be transmitted across the communications media.

The use of the ATNPkt for authentication is illustrated in Figure 13 and is detailed further in this paper. The IPS gateway will not have any UDP ports other than 5908 with a key tag of 0x0A available for unauthenticated aircraft. All messages in the authentication sequence will have UDP port 5908 and the first byte of the UDP data field will have a key tag value of 0x0A preceding the ATNPkt. During authentication, the ATNPkt carries the DTLS data in the user data. After the DTLS Logon handshaking is complete the avionics will send an additional encrypted DTLS application packet with the aircraft's IP address, tail number, ATN Address, Random Sequence Number and Flight ID for addressing use by the IPS gateway. The IPS Gateway will respond with an encrypted DTLS application packet including a Random Sequence number. After

authentication has been completed, anything on port 5908 with a key tag of 0x0A will be TLS Alert messages and/or connection maintenance traffic (note after DTLS session all messages include MIC). The Sequence number is used in the Message integrity check formula.

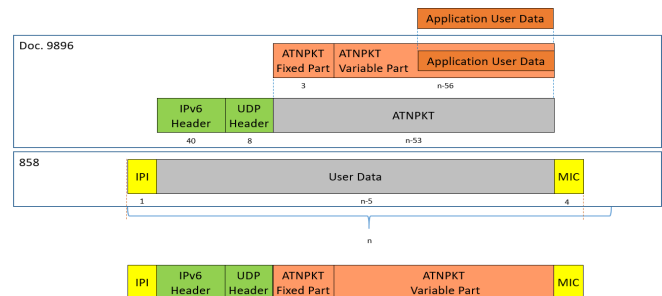


Figure 11. IPS Authentication Protocol Build Up

DTLS is an enhancement on TLS for secure UDP connections. The DTLS Protocol is recorded in RFC 6347 [DTLS]. There are six flights to a DTLS session establishment for the secure connection, shown in Figure 14. During the initial rollout of IPS, a modification on the TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 method will be used. Crypto scientists have determined that SHA256 is near the end of usefulness so an upgrade to SHA384 will be used [36], [37]. To facilitate maximizing the utilization of packets, the Deflate compression option already built into DTLS will be used.

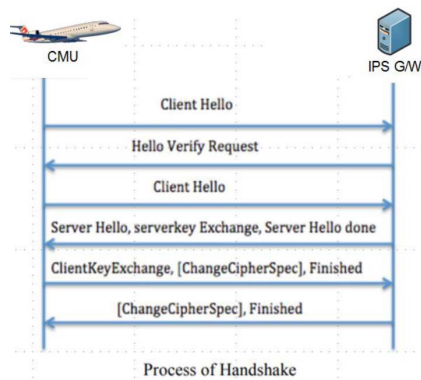


Figure 12. DTLS Logon Flights

ECDSA keys pairs will be provided by the primary service provider for each aircraft subscribed to the IPS service. The keys will be signed by the primary service provider's own or designates CA key and be verifiable by any entity possessing the service provider's or designates public key. (A trusted companion service provider) For example: If ARINC was the service provider for American Airlines (AA) and a AA aircraft was operating in China, it would be able to authenticate with ADCC if ADCC possessed a copy of ARINC's or designate's Root Certificate.

Each aircraft will receive two public certificates and two private keys. The public certificate is used for authentication with the IPS Gateway(s) and the private key is kept secret with the aircraft. Each undoes the encryption of the other and must work in pairs to establish and maintain secured connections.

To minimize the size of the public keys, they will be encoded in X.509 certificate DER format [27]. The private keys are never transmitted in an authentication exchange. Each key's valid dates will correspond with existing contract dates plus a grace period if applicable between the airline and the primary service provider.

In the event that an aircraft key is compromised, the aircraft will have a one-time-use backup key that can be used for authentication. This back-up key will only be valid on the primary service provider's network to facilitate upload of replacement keys. After using a backup certificate, if new keys are not uploaded the airline must data-load new certificates and keys. The Avionics will support a way to replace the existing public keys and certificates using both a physical media and also over the air.

7. PKI Certificate Management

Each X.509 certificate has parameters that identify the valid user of the certificate. Certificates will include the aircraft's public key, a signed hash using the service provider's private key, and the following additional information as listed in table 1.

7.1 X.509 Certificate Parameter for Non-Aircraft

Maintenance devices may require certificates, which give permission for the generation of Certificate Signing Requests (CSR) for a particular airline and primary service provider. Having certificates on the maintenance device(s) would allow that device to create CSRs for one particular airline, and service provider. Devices could then be kept secured to ensure that only authorized people and avionics receive valid certificates thus preventing unauthorized people from installing billable certificates on unauthorized avionics. The Certificate Policy and Certificate Practice Statement will expand on this concept further.

Table 1. X.509 Certificate Parameters for Aircraft

Field	Value	Example
Country Name [AU]:	2 letter country code of airline host	US
State or Province Name	Full Province or state name of airline host	Georgia
Locality Name	City of airline host	Atlanta
Organization Name	issuing airline	Delta Airlines
Organizational Unit Name	ICAO Airline Designator	DAL
Common Name	Tail Number. Aircraft Type. ICAO Code. Service	N12345.A380.DAL.IPS

Field	Value	Example
Email Address []:	PKI Sponsor E-mail	PKI@delta.com
A challenge password []:		[None]
An optional company name		[None]
Issuer	Service providers information	Rockwell Collins ARINC NA
Validity	Dates and time period key is valid	[Contract specific]

7.2 X.509 Certificate List

It shall be the responsibility of each service provider or designate to maintain a service key directory of X.509 certificates for all aircraft for which they are the primary service provider. It also shall be the responsibility of each primary service provider to maintain a valid public CA X.509 certificate in DER encoding with all other trusted companion service providers for which a trusted relationship is established.

7.3. Service Provider Trust Relationships

Each service provider shall have the option to enter into roaming agreements with other service providers. These trusted roaming providers shall be called trusted companion service providers. If a companion service provider has a valid trust operating agreement then an exchange of public root CA certificates between providers or the establishment of a trust bridge will allow aircraft to utilize the companion network while in transit. Certificates shall be encoded in DER format.

7.4. Aircraft Roaming and Keys

It is up to each airline to determine which service providers they wish to allow their aircraft to connect with if any. This is bounded by the trust relationships between service providers. If a set of trusted service providers are desired, the aircraft avionics should be loaded with server certificates for each trusted service provider. The aircraft will then

be able to authenticate the IPS Gateway and the IPS Gateway will be able to authenticate the aircraft.

By way of example if ADCC and SITA enter into a trusted relationship: Aircraft that have ADCC as their primary service provider will have the option to roam onto the SITA network, if the aircraft is equipped with SITA's gateway server certificate. Without this trusted relationship then aircraft will not be able to roam onto the other's network even if the avionics contained the SITA certificate. In this case the SITA IPS Gateway would reject aircraft presenting a certificate signed by ADCC.

Avionics should disable IPS if they do not at a minimum have an Aircraft Public Certificate, Aircraft Private Key, Primary Service Provider's Public Server Certificate and a Primary Service Provider's CA Certificate(s). Having a Onetime Use key and certificate is highly encouraged to recover aircraft whose keys expired while out of the primary service provider's area.

Assuming the aircraft is roaming onto another service provider's network area. The following truth table depicts in Figures 15 and 16 whether the aircraft will accept or reject the Trusted Companion service provider's server key.

Service Provider Aircraft key store	Has Trusted Companion Public Certificate	Does not Have Trusted Companion Public Certificate for Aircraft's Primary Service Provider.
	Key store	
Has Secondary Service Provider Server Key	Server Key accepted – Logon continues	Ground issues a DTLS Alert message and discontinues the connection.
Does not have new Service Provider's server Key	Aircraft discontinues communication with this service provider. Aircraft may issue a DTLS Alert message	Ground issues a DTLS Alert message and discontinues the connection.

Figure 13. Avionics Login Results Table (Trusted Service Provider)

Service Provider Key store	Has Primary Service Provider Server Public Certificate
Aircraft key store	
Has primary service provider Server Key	Server Key accepted – Logon continues
Does not have primary service provider's server key	Misconfigured Aircraft cannot authenticate with Primary Service Provider

Figure 14. Truth Table Logon Results (Primary Service Provider)

7.5. Key Revocation List - CRLs

Each primary service provider shall maintain a certificate revocation list. Any key generated by the primary service provider that is later compromised, other than by expiration shall be listed in a certificate revocation list until the certificate expires. This list is to be shared no less than daily with all trusted companion service providers, even if no changes are recorded. It is recommended that an encrypted method be established for sharing these lists.

One time use keys may be distributed to trusted companion service providers as a Certificate Revocation list as well. Each Aircraft will be equipped with a one-time use certificate from its primary service provider. These certificates will be included in CRL lists provided to trusted companion providers, effectively making these certificates one time use only on the primary service provider's network. In the event that the aircraft's primary certificate fails due to expiration or CRL revocation the aircraft can use this one-time use key on the primary service provider's network. The one time use key will expire upon first use. Having a one-time use key ensures that aircraft will not require physical media in order to replace its service keys. That is as long as it is connected with the primary service provider. Uploaded one-time use certificates will be in DER format and be via the DTLS encrypted channel negotiated at the time of authentication session establishment.

Aircraft should maintain their DTLS connection with the service provider after installing a new one time use certificate using the old certificate if necessary. The UDP port 5908 key tag of 0x34 will be followed by a one-time use certificate in DER format when sent by the Service Provider. The

aircraft will use the port 5908 key tag of 0x34 and one additional byte to indicate success or failure.

Online Certificate Status protocol is recommended between trusted service companions but not required. It will be up to each service provider to setup how it wants to interact with other trusted service providers. OSCP availability does not alleviate the need to publish CRLs to trusted companion service providers. OSCP is seen as a useful resource but not impervious to outages due to network connectivity issues and server hardware failures.

7.6. Elliptic Curves

To simplify the authentication exchange and session key generation a named pre-configured elliptic curve generally accepted by the security community will be used. Initially the curve will be secp384r1, however future support for secp521r1 is expected.

7.7. Encryption

AES 256 will be used for encrypting all message traffic on UDP port 5908 with a key tag of 0x0A, or 0x3X after authentication is complete and during any key maintenance operations. All other traffic on this and all other ports will be sent unencrypted; however a Message Integrity Code (MIC) will be generated to ensure the message was not tampered with while in transit.

7.8. Hash

Initially the hashing function shall be the same for the MIC as that used on the client's/air craft's ECDSA Keys. The Hashing function for MIC generation will be negotiated during the authentication process. SHA 384 hashing algorithm is selected for MIC generation. All but the last 4 Bytes will be truncated to minimize the length of the hash while maintaining the security value.

7.9. Message Integrity Check

The message integrity check (MIC) is computed for each IPv6 packet, for non-IP networks the MIC may also be computed for each subnetwork packet transmitted in order to secure the subnetwork (this is the case for VDL Mode 2, other subnetworks may be

different). The MIC is computed after authentication has been completed.

7.10. MIC for IP Packet

The MIC is computed for each IPv6 packet. A fragmented application message, consisting of a number of IPv6 packets, will have a MIC on each packet. The MIC is computed after compression over the entire IPv6 packet, the scope of the MIC computation is shown in Figure 17. The last 4 bytes from the MIC computation are used to populate the MIC field, which is added at the end of the IPv6 packet by the IPS Gateway for uplink messages. For downlink messages, the IPS Gateway computes the MIC the same way and compares the last 4 bytes against the value in the MIC field received in the downlink message. If the values do not match, the message is logged with the status of invalid MIC and a DTLS alert message (bad_record_mac) is generated in response.

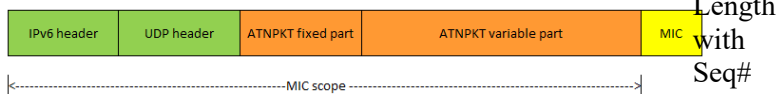


Figure 15. MIC Scope for IP Packet

7.11 Message Integrity Code (MIC) Generation Function

DTLS uses the following function to generate the message integrity code:

$$MIC = Truncate(4, PRF(App\ Data + Seq\# + Data\ Length\ with\ Seq\#, + Session\ Key + Key\ Length))$$

“+” denotes concatenation. The MIC is generated before any encryption is applied. If encryption is applied, it includes the MIC.

Variable Name	Explanation
Truncate	A Truncate function that reduces the size of the operator to a number of bytes. In this case the last 4 bytes of the message hash function will be used as a message integrity check
PRF	Pseudo Random Function: This is the hashing function negotiated during the initial DTLS handshake. The initial supported hash will be SHA-384.
App Data	The application layer of data including the header to be MIC'd.

Variable Name	Explanation
Seq #	The random sequence number sent in the last message after authentication added with the total transmissions since that time. This Seq# is unique for downlinks and uplinks and starts with a random number sent after successful DTLS logon. For example the downlink sequence number could be 568 and the uplink sequence number could be 123. After one downlink the new downlink sequence number will be 569. Unlike TCP there is one set of sequence numbers for aircraft communications rather than by stream. The Sequence number rolls over to zero if it reaches it max. This sequence number is not to be confused with the Orange sequence number, if any.
Data Length with Seq#	The total length of the Application Data added to the current sequence number. If the results is greater than max value. Subtract max value. This is effectively a check on the data integrity.
Session Key	This is the lower 32 bytes of the session key derived as per RFC 5246 Section 6.3. Both the gateway (server) and aircraft (Client) have a session or master key and compute the counter parties key using the procedure recorded in the RFC. This value is never transmitted making the PRF function difficult to duplicate by third parties.
Key Length	The total session key length in bytes.

8. Two-Way DTLS Based Secure Aircraft Communication

Our proposed security scheme is therefore based on EEC, the efficient, matured and widely accepted public key cryptography algorithm. It is designed to work over standard communication stacks that offer UDP/IPv6 networking for ATN/IPS. Our implementation of DTLS is presented in the context of a system architecture and the scheme’s feasibility (low overheads and high interoperability) is further demonstrated through extensive evaluation on a

hardware platform suitable for the aviation safety service air to ground communication.

8.1 Proof of Concept

In order to validate the proposed protocol, we implemented a prototype of the delegation-based key provisioning protocol. The discussion includes the framework design, implementations choices as well example results from the implementation demonstration.

Modularized prototype framework was designed to authenticate aircraft and ground entities. Using DTLS and crypto library, aircraft and ground provide the security to protect air to ground communication. Aircraft initiate the authentication on each access network with the ground peer. After the mutual authentication ground and aircraft exchange the secret key and generate session keys which will be used to encrypt, decrypt and message integrity check the messages.

As one party may open more than one protocol session related to different peer parties, the Session Manager is introduced to coordinate different sessions and handle related operations. We also develop the message manager which provides message Encoding/Decoding and also provides an API for the Message module to parse and construct messages. We developed the crypto library wrapper, which wraps the underlying cryptography library and provides cryptographic computation APIs for the message and other modules.

8.2 Implementation Choice and Considerations

As for the underlying crypto library, we investigated various crypto libraries. For the avionics side we investigated the non-optimized version of crypto library that is being used in the various Rockwell Collins products, wolfSSL and a few others. For the ground side we selected the Openssl. Openssl is feature rich, widely used and advanced. At the point of implementing the protocol prototype, to the best of our knowledge, we found a few bugs in Openssl. Openssl is more stable and easier to use with respect to the cryptographic algorithms used in ground side prototype. On the avionics protocol prototype, we adopted our implementation of crypto library as the underlying crypto library and its

wrapper as crypto/networking APIs to facilitate the crypto and networking operations.

In the protocol specifications, authenticated encryption is used to protect air to ground messages carrying air traffic and AOC messages. The prototype adopts AES Galois/Counter Mode (GCM) [4] as the authenticated encryption algorithm, which is a block cipher with both encryption and authentication functionalities. With AES-GCM, the key material Y is first encrypted and then authenticated together with the session information to generate an authentication tag. As a result, the token r and p consist of three parts: a plaintext of the session information, a ciphertext of the key material and an authentication tag. The session management module has the responsibility of monitoring and checking the session information. For example, in a new session, when a party receives the fresh session information, namely relevant identifiers and nonces, from an incoming message, it should first validate the freshness and the correctness of the session information. Our implementation choices for the cryptography algorithms is listed in Table 2.

Table 2. DTLS Session Parameters

Field	Value
Keys	ECDSA
Diffe Hellman	ECDHE
Elliptic Curve	Secp384r1
Encryption	AES 256 GCM
Hash	SHA-384
	ECDSA

9. Data Analysis and Result

As an illustration, we obtained the following results for the aircraft and the ground (see Table 3 based on the setup shown in Figures 2 and 4). Session is establish between message gateway and avionics. Message is exchanged between avionics and an ANSP IP end systems as shown in Figure 4. Each message is sent with a 5 second gap in between. We repeated the test 75 times and we observed that the time for the DTLS session execution is around 4.471sec (we suppose that the network latency is the same for both DTLS with and without compression and therefore it is ignored by our comparison). Credential protection adds an extra cost of about 0.23sec, which is induced by two additional cryptographic operations: the calculation of a 256 bits

encryption key using the PRF function, and the ECC encryption/decryption of the aircraft certificate.

The session establishment time shown in Table 3 is a measure of the handshake processing cost. DTLS session establishment require key exchanges data transfer and session key and total time for the session key establishment for the secure session is reasonable and workable for the resources constraints and bandwidth limited air to ground links. However, we believe the longer processing cost is overstated because no attempt was made to use the optimized versions of the crypto libraries, which boast of performance improvement.

Table 3. DTLS Session Parameters

Description	Key Exchanged for Mutual Authentication, crypto	Execution Time (seconds)
DTLS Protection	ECDHE-ECC	4.471
Encrypted payload (aircraft to ground)	AES-GCM	0.451
Encrypted payload (ground to aircraft)	AES-GCM	0.401

10. Conclusion

In this paper, we focused in depth on the security aspects of the various avionics architectures. Starting with a general security analysis, we identified security and privacy concerns unique to avionics systems. Then we proposed a unified avionics architecture design based on the ground IPS network that addresses security concerns and increases the confidence in the assured operation of IPS systems. We introduced a new layer in the architecture, which we refer to as the IPS gateway that connects heterogeneous network hardware in airborne ATN/IPS systems to the global aviation IP network. We presented a novel and efficient approach to providing strong security for aviation safety services using Internet protocol IPv6 and UDP communications in aviation networks. By extending the industry standard protocol for securing UDP communications, called DTLS, and integrating an ECC based public key cryptography, ATN/IPS achieves security in the existing low bandwidth air to ground links while eliminating some overhead but

also adding some overhead associated with mutual authentication and certificate management while still providing proven security services to UDP communications. Throughout the paper, we have discussed areas of further research in air to ground security, the current DTLS implementation should be enhanced to support multiple, concurrent sessions. We are currently investigating efficient implementation of our approach in embedded system to support multiple applications and plan on collecting real world performance data during the validation by performing laboratory as well as aircraft flight tests. Our efforts extend the Internet Protocol to include the real world objects and therefore facilitates that are the formation of a large and new network inclusive of avionics devices.

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